Roll No.....

End Semester Examination

Name of the Program: B.Tech(ECE) Semester: IV
Name of the Course: Communication Systems I Course Code: TEC 401

Time: 3 Hours Maximum Marks: 100

Note:
(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question
(iii) Total marks in each main question are twenty
(iv) Each question carries 10 marks.

Q1	(20 marks)	
(a)	Explain the modulation and demodulation of VSB wave using filter method with the help of neat block diagram.	
(b)	(i) Show that a signal and its Hilbert transform have the same autocorrelation function. (ii) SSB is suitable for speech signals and not for video signals. Why?	**CO1**
(c)	Derive an expression for the effective modulation index of a multi-tone amplitude modulated signal.	
Q2	(20 marks)	
(a)	Derive an expression for a single tone narrowband frequency modulated wave.	
(b)	Explain, with the help of block diagram, the Armstrong method of generating FM signal.	
(c)	An angle modulated signal is described by $S(t) = 5\cos[2\pi 10^6 t + 200\sin(10^3 \pi t)]$. (i) Considering $S(t)$ as a PM signal with $k_p = 100$ rad/volt, find the modulating signal $m(t)$. (ii) Considering $S(t)$ as an FM signal with $k_f = 10^5$ Hz/volt, find the modulating signal $m(t)$.	**CO2**
Q3	(20 marks)	
(a)	Consider an angle modulated signal $S(t) = 10\cos[\omega_c t + 3\sin(\omega_m t)]$. Assuming PM and $f_m = 1KHz$. Calculate the modulation index and the bandwidth when (i) f_m is doubled and (ii) f_m is decreased by one half.	**CO 2 & 3**
(b)	Derive an expression for the overall equivalent noise temperature for cascaded stages of amplifiers.	
(c)	Prove that the root mean square noise voltage developed across the capacitor in a parallel R-C circuit is given as $\sqrt{\frac{kT}{C}}$, where k is the Boltzmann's constant and T is the absolute temperature.	
Q4	(20 marks)	
(a)	What is the necessity of pre-emphasis and de-emphasis used in FM system? Explain pre-emphasis and de-emphasis with neat circuit.	**CO4**
(b)	Assuming sinusoidal modulation, show that in an AM system with envelope detection the	

	figure of merit is given by $\gamma_{AM} = \frac{\mu^2}{2+\mu^2}$, where μ is the modulation index for AM	
(c)	Show that in an FM system the figure of merit, assuming sinusoidal modulation, is given By $\gamma_{FM} = \frac{3}{2} m_f^2$, where m_f is the modulation index for FM	
Q5	(20 marks)	**CO5 & CO6**
(a)	Draw the block diagram of a communication system and explain the function of each block.	
(b)	With the help of neat circuit diagram, explain the generation and demodulation of PWM.	
(c)	State and prove sampling theorem.	